

Summary

Data Engineer with expertise in building and optimizing data pipelines using Python, SQL, and PySpark. Proficient in large-scale data processing with Apache Spark, Azure Data Factory, Microsoft Fabric, and Databricks. Skilled in modern data warehousing architectures and Delta Lake/Delta format for efficient, reliable storage and analytics. Successfully improved data-processing speed by 25% and automated workflows on Azure, reducing manual intervention by 40%. Adept at designing and optimizing ETL/ELT pipelines to enhance performance and scalability, and eager to leverage cloud platforms to strengthen data infrastructure and advanced analytics capabilities.

Education

University at Buffalo, The State University of New York (Jan 2024 – May 2025)

Masters in Data Science

Channabasaveshwara Institute of Technology (Jun 2012 – May 2016)

Bachelors in Computer Science

Experience

Senior Data Engineer, Sogeti, USA

Aug 2025– Present

- Designed and implemented scalable data pipelines using **Databricks** to process structured and semi-structured data from diverse sources.
- Developed data transformation logic in Databricks **notebooks** to cleanse, aggregate, and enrich raw data for downstream analytics and ML use cases.
- Utilized **Delta Lake** for versioned, **ACID**-compliant data storage with time travel and data auditing capabilities.
- Developed workflows for Pyspark and **Spark SQL** notebooks, with scheduled triggers for execution.
- Implemented **Databricks Unity Catalog** to centralize data governance for 5TB of sensitive data, enhancing data lineage tracking by 60% and supporting consistent change management via **CI/CD** pipelines in **Azure DevOps**.
- Implemented **RBAC & ABAC** rules using Unity catalog to protect sensitive information within the Databricks environment.
- Automated ingestion of large-scale datasets using **Auto Loader** with **cloudFiles API**, ensuring real-time availability of fresh data for downstream analytics and machine learning pipelines.
- Wrote queries to create **stored procedures**, User Defined Functions, Views **CTE** for Data loading and transformation into SQL Warehouse.
- Implemented **incremental loading** strategies using watermarking and delta merges in Databricks to support near real-time data processing.
- Performance tuning of multiple databases by fixing **indexes**, table **partitioning**, usage of temp tables, CTE redesigning schemas, and tuning indexes (clustered, non-clustered).
- Strong knowledge of **data warehousing** concepts, including **fact and dimension** tables, **star/snowflake schema** modeling, and hands-on experience implementing dimensional models and **SCD Type 1 and Type 2** in SQL Server.
- Optimized Synapse queries with materialized views, partitioning, and distribution strategies to ensure quick response for critical dashboards.
- Implemented **RBAC** and **Synapse** to secure sensitive PII data in compliance with data governance standards.
- Designed and deployed scalable, secure cloud data architectures on **Microsoft Fabric** using **Lakehouse**, Data Warehouse, and **OneLake** to integrate on-premises and multi-cloud sources for enterprise analytics.
- Developed and optimized end-to-end ingestion and transformation pipelines with Data Factory and **Dataflows Gen2**, storing curated data in Delta Lake tables to boost processing efficiency and query performance.
- Implemented **data quality checks**, validation rules, and monitoring mechanisms to ensure accuracy and consistency across **Fabric Lakehouse** datasets.
- Optimized** query and storage performance by applying partitioning, indexing, and efficient file management techniques within **Fabric**, improving downstream analytics performance
- Led **migration** initiatives from **Snowflake to Databricks**, including assessment of existing data models, redesigning transformations using Spark/Delta Lake, migrating data to cloud object storage, validating performance benchmarks, and implementing governance and access controls in the Databricks environment.
- Re-engineered Snowflake SQL transformations and tasks into **Spark-based pipelines**, optimizing performance and enabling distributed processing.
- Performed **data validation**, reconciliation, and performance benchmarking to ensure data accuracy, query parity, and improved workload efficiency post-migration.
- Implemented **governance** and **security** controls in Databricks, including Unity Catalog, RBAC, and cluster policies to maintain compliance and access control standards.
- Optimized** compute costs by transitioning from warehouse-based compute (Snowflake) to auto-scaling clusters and **job clusters** in Databricks.
- Led and **mentored** a cross-functional **offshore team**, conducting code reviews, setting technical standards, and guiding best practices across Snowflake, Databricks, and ETL development.
- Coordinated **sprint planning**, **task allocation**, and **stakeholder communication**, ensuring on-time delivery of data migration and analytics initiatives while fostering collaboration and continuous skill development within the team.

Senior Data Engineer, Deloitte, India

Apr 2021 – Dec 2023

- Developed and implemented data-driven forecasting models and **ETL/ELT pipelines** to clean, transform, and aggregate data, enhancing data accuracy and supporting data governance practices to meet stringent quality standards.
- Designed and automated **Azure Data Factory** pipelines for sales and distribution data processing, improving data movement efficiency by 32% and supporting rigorous change management and automated reporting.
- Optimized complex **SQL, T-SQL** queries and **PySpark** workflows in **Databricks**, processing large-scale data and applying scalable data quality validations and business rule checks.
- **Optimized** data warehouse solutions in **Azure Synapse Analytics** using **dimensional modeling** and medallion architecture, increasing query performance by 40% and reducing retrieval time by 30%, while strengthening security and compliance via **Azure Key Vault**.
- Developed scalable, Azure-based data solutions to streamline production and inventory tracking, reducing project delivery timelines by 30% while mentoring junior analysts on data management best practices.
- Collected data using **Spark Streaming** from **Azure Data Lake Storage** in near-real-time and performs necessary Transformations and Aggregation on the fly to build the common learner data model and persisted the data in **Delta Lake** to support real-time analytics workloads
- Optimized data lake performance by implementing **Delta Lake** on **Azure Databricks**, enabling ACID transactions and improving data reliability, resulting in a 30% reduction in data reconciliation efforts.
- Implemented **Change Data Capture** (CDC) using Azure Data Factory and SQL Server to incrementally process only modified records, significantly reducing pipeline run times and improving data freshness for real-time reporting dashboards.
- Built real-time ingestion pipelines using **Databricks Autoloader** and **Delta Live Tables** to stream high-velocity data from **Azure Event Hubs**, improving pipeline reliability by 70% for downstream analytics workloads.
- Optimized existing algorithms in Databricks by leveraging Spark Context, **Spark SQL**, **DataFrames**, and Pair **RDDs** to enhance performance. Processed both schema-oriented and schema-less data using Apache Spark.
- Implemented **Microsoft Fabric** with **OneLake** for unified data storage and analytics, ensuring data lineage and consistent data governance across large datasets.
- Designed and maintained **Lakehouse** architecture using Microsoft Fabric and OneLake to unify structured and unstructured data, enabling seamless collaboration between data engineering and analytics teams.

Data Engineer, Capgemini, India

Nov 2019 – Mar 2021

- Enhanced data models with **SQL** and **PySpark** to improve customer satisfaction by 22% and support robust data analytics, aligning with automated reporting and governance best practices.
- Developed custom UDFs in Python to be used in spark-SQL.
- Built and orchestrated Data Factory pipelines to ingest and transform data from diverse sources including **REST APIs**, **structured and semi-structured** files (CSV, JSON, Parquet), **Cosmos DB**, and **PostgreSQL**, ensuring data integrity, scalability, and streamlined change management for high-quality analytical solutions.
- Delivered actionable insights from large datasets using **Power BI**, developing clear documentation and robust data validation methods to support enterprise-level data governance and technical documentation.
- Implemented robust **incremental load** strategies using watermarking and delta logic in Azure Data Factory, optimizing pipeline performance and reducing data processing time by over 40%.
- Integrated **Azure Logic Apps** with ADF pipelines to trigger alerts and automated workflows based on data quality thresholds and job status, improving system transparency and incident response time.
- Collaborated with business partners, Business Analysts, and Product Owners to gather requirements and develop scalable, distributed data solutions using the Hadoop ecosystem on Azure HDInsight.
- Implemented robust batch processing workflows to transform and aggregate large datasets on a regular schedule, ensuring timely data availability for downstream **analytics** and **reporting**.
- Skilled in **Azure SQL Data Warehouse** (Azure Synapse) for enterprise scale reporting and analytics, and **Azure Databricks** for efficient data ingestion, transformation, and processing
- Hands-on experience with workflow orchestration tools (Azure Data Factory, Airflow) and automation utilities (SFTP, Logic Apps) to streamline batch and streaming processes.
- Performed database performance tuning by analyzing execution plans, optimizing indexes, and triaging query bottlenecks in SQL Server, MySQL, and Synapse.
- Managed pipeline monitoring, logging, and alerting, and automated error handling and retry mechanisms.
- Scheduled and triggered pipelines using **ADF triggers**, Event-based, and Time-based scheduling.
- Created and managed external tables in Azure Synapse to efficiently query data stored in Azure Data Lake Storage Gen2, enabling seamless data virtualization and reducing data duplication across environments.
- Led **GitHub** code review sessions to refine code deployment processes in a fast-paced **Agile** environment, reducing post-deployment issues by 28% and reinforcing SDLC best practices.

Data Engineer, DXC Technology, India

Feb 2017 – Oct 2019

- Tuned **MySQL** database performance, leading to a 16% rise in daily active users on the customer-facing web portal by reducing page load times and query latency
- Successfully completed a **POC** for Azure implementation, with the larger goal of migrating on-premises servers and data to the cloud. Involved in various phases of **Software Development Lifecycle** (SDLC) of the application, like gathering requirements,

- design, development, deployment, and analysis of the application.
- Developed ETL pipelines using Azure services to migrate data from on-premise MySQL databases to cloud-based storage, using SQL scripts to extract, cleanse, and load data into staging layers, supporting a phased cloud modernization initiative.
 - Worked on SQL and PL/SQL for backend data transactions and validations
 - Participated in **root cause** analysis of data issues by tracing data lineage across multiple systems, helping reduce recurring data inconsistencies by 40%
 - Experienced with ITIL processes and **ServiceNow** for incident, change, and problem management to ensure smooth production support and compliance.
 - Assisted in the **UAT** process by preparing test data and validating report outputs, contributing to a 98% success rate in production deployment

Skills

Skill Area	Key Skills
Data Storage	Data Lakes, Data Warehousing, SQL & NoSQL Databases, Cloud Storage
Data Processing	ETL/ELT Development, Data Pipelines, Batch & Stream Processing, Data Transformation
Big Data Technologies	Hadoop, Spark, Distributed Computing, Large-scale Data Processing, Databricks
Cloud Platforms	Azure (Azure Data Factory, Azure Synapse Analytics, Azure Databricks), Microsoft Fabric, Snowflake
Programming Languages	Python, SQL, PySpark
ETL/ELT Tools	Azure Data Factory
Analytics & Reporting	Data Visualization, BI Tools (Tableau, Power BI), Descriptive & Predictive Analytics
Orchestration	Workflow Automation, Job Scheduling, Data Pipeline Orchestration, Kafka
Machine Learning	Model Deployment, Feature Engineering, Integration with Data Pipelines
DevOps & CI/CD	Version Control (Git), Continuous Integration/Continuous Deployment (CI/CD)
Networking	Data Integration, API Management, Cloud Networking, Data Migration
Data Quality	Data Profiling, Data Cleansing, Data Validation, Data Lineage
Version controls and Tools	GIT, Azure DevOps
Data Warehousing	OLAP, Dimensional Modeling, Schema Design, Data Marts, Snowflake

Certifications

- Microsoft Certified Azure Data Engineer Associate
- Microsoft Certified Fabric Data Engineer Associate

Achievements

- Received Shining Star, Spot, and Applause awards in recognition of exceptional contributions to data modeling initiatives that enabled deeper financial insights and improved decision-making across business units.